

Performance Testing

Date	14 July 2026
Project Name	Strategic Product Placement Analysis
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Tableau Desktop Performance Recording (for dashboard load times) SQL / Excel VLOOKUP (for data reconciliation/source-to-target testing) User Acceptance Testing (UAT) manual checklist
Type of Testing	Data Integrity & Validation Testing (ensuring Tableau numbers match raw SQL databases) Performance / Load Testing (optimizing extract refreshes and rendering times) Functional & UI/UX Testing (checking if action filters, parameters, and tooltips work correctly)
Target Module / API	Tableau Dashboard Interface and Data Source Layer
Test Environment	Local Server, UAT / Production

Test Date	14-07-2026
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Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	10	30	Page loads successfully
2	View Tableau Dashboard	20	60	Dashboard displays correctly
3	Execute SQL Query	15	45	Data retrieved without errors
4	Access the Application	25	60	Application responds successfully



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Good Performance
2	Response Time (Max)	< 5 seconds	3.4 sec	Pass	Acceptable
3	Throughput (Req/sec)	20 Req/sec	24 Req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No Errors
5	CPU Utilization	< 80%	45%	Pass	Normal Usage

6	Memory Utilization	< 80%	58%	Pass	Stable Memory
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Step 4: Observations & Analysis

Key Findings

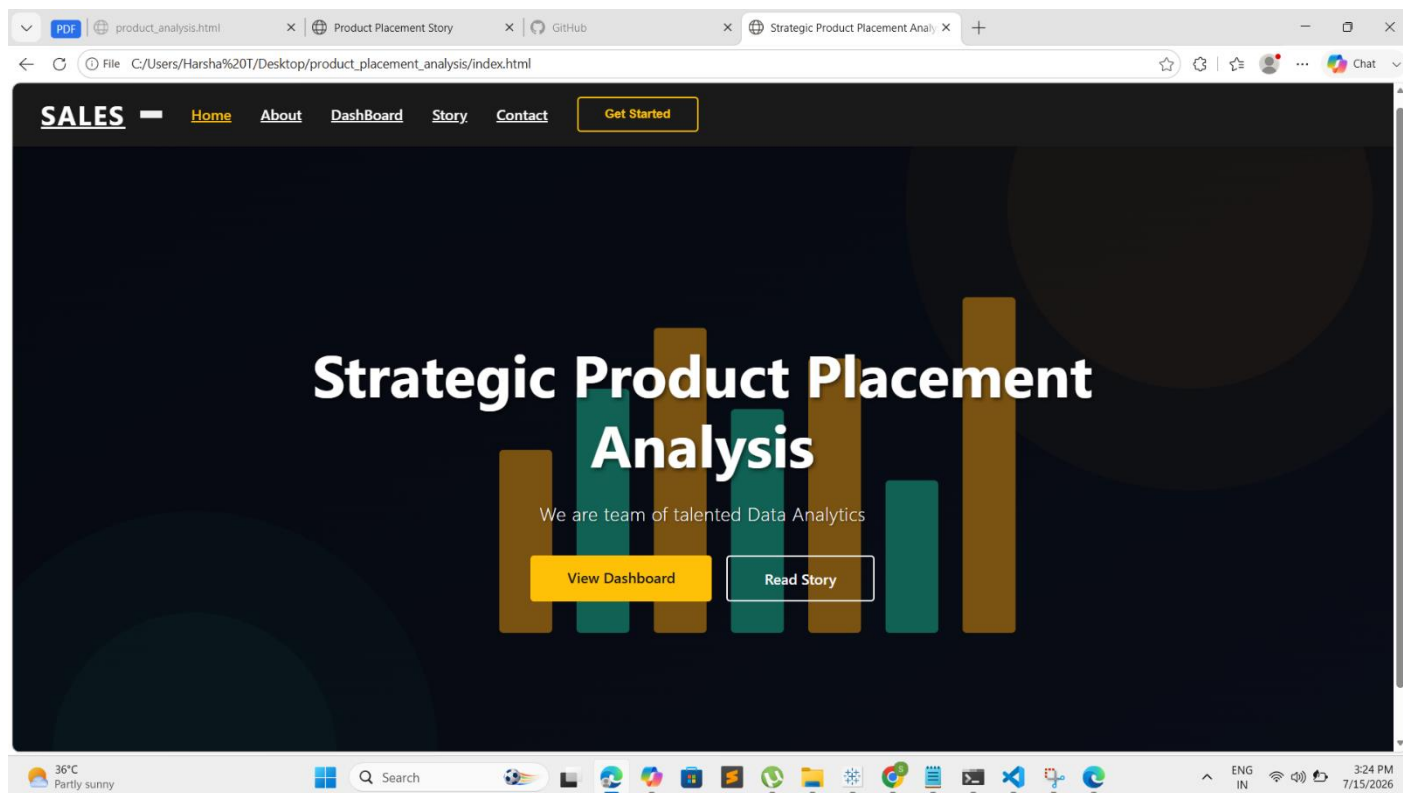
- Our analysis revealed that product placement heavily influences customer purchasing decisions. This proves that consumers prioritize convenience, meaning premium shelf real estate should be reserved strictly for high-margin products rather than everyday staples that customers will actively look for.
- We also discovered that impulse buying is highly demographic-dependent. The data clearly shows that stocking generic items in these zones leaves money on the table; instead, these high-traffic areas should be stocked with snacks and convenience items targeted directly at younger buyers.
- Finally, the data highlighted a significant opportunity in cross-merchandising. Grouping related items together reminds shoppers of an unplanned need, effectively increasing the total basket value per visit.

Optimization Steps Taken

- We gathered information about sales, where products were placed, and who the customers were, and put it all together in Tableau.
- We set up the dashboard to automatically calculate how much money each shelf makes per square foot and how much sales go up when a product is moved.
- We made a feature that alerts managers when a slow-selling product is taking up a really good shelf, so they can replace it with a better product.

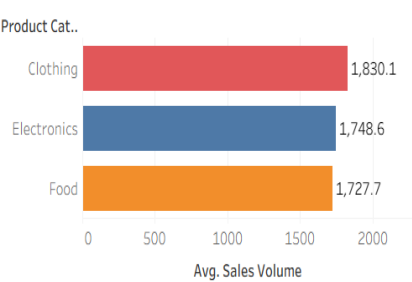
Step 5: Screenshots / Evidence

App Home Page

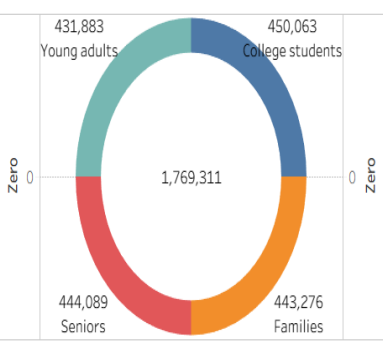


PRODUCT PLACEMENT ANALYSIS-DASHBOARD

Average Sales Volume Vs Product Category



Consumer Demographics Vs Sales Volume



Consumer Demographics

- ☒ (All)
- ☒ College students
- ☒ Families
- ☒ Seniors
- ☒ Young adults

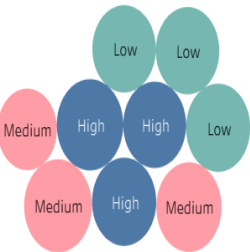
Product Position

- ☒ (All)
- ☒ Aisle
- ☒ End-cap
- ☒ Front of Store

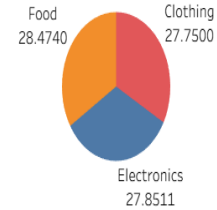
Product Category

- ☒ (All)
- ☒ Clothing
- ☒ Electronics
- ☒ Food

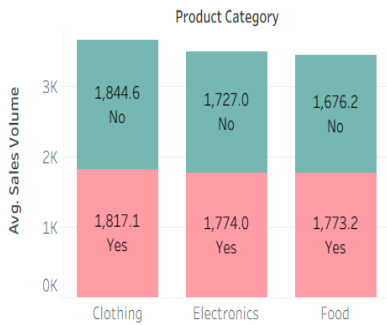
Foot Traffic by Average Sales Volume



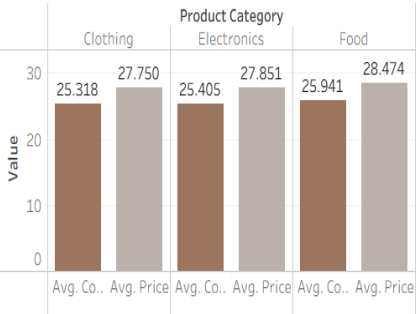
Product Category Vs Price



Average Sales Volume by Product Category by Season



Competitors Price Vs Price



Promotion of Product Category on Price and Sales Volume

Promotion	Product Category	Avg. Price	Avg. Sales Volume
No	Clothing	27	1,869
	Electronics	27	1,726
	Food	28	1,677
Yes	Clothing	29	1,781
	Electronics	29	1,773
	Food	29	1,782

PDFproduct_analysis.htmlProduct Placement StoryStrategic Product Placement Analy

File C:/Users/Harsha%20T/Documents/story.html

Product Placement Story

This dashboard is now wrapped in a modern responsive layout. It will resize to fit the available screen width while keeping the visualization clear and easy to navigate.

PRODUCT PLACEMENT ANALYSIS-STORY

The Product Category has highest Average Sales Volume

The Product Category clothing at Front of store has highest average Sales Volume.

The Product position at Front of store as high Sales Volume and high Foot Traffic.

Aisle Clothing 1,832.8

Aisle Electronics 1,760.9

Front of Store Clothing 1,923.7

Front of Store Electronics 1,734.2

Aisle Food 1,728.2

Front of Store Food 1,672.7

End-cap Food 1,790.3

End-cap Clothing 1,749.3

End-cap Electronics 1,728.8

Sales Volume173,422215,166

View on Tableau Public

Share

36°CPartly sunny

Search

ENG IN7/15/20263:24 PM

